

Question-1: What is preference based approach of consumer's behavior? Explain and interpret the assumptions of preference based approach (e.g., Completeness, Reflexivity, Rationality).

Let's begin our analysis of individual decision-making in an abstract setting.

• Let $X \subseteq \mathbb{R}_+^N$ be a set of possible alternatives for a particular decision maker.

→ It might include the consumption bundles that an individual is considering to buy.

Example: $X = \{x, y, z, \dots\}$

or, $X = \{\text{Apple, orange, banana, } \dots\}$

There are two ways to approach the decision making process:

1. Preference-based approach.

2. Choice-based approach.

Preference-based Approach:

It analyses how the individual uses his preference to choose an element(s) from the set of alternatives X .

Preference-based approach is more tractable when the set of alternatives X has many elements.

'Attitudes' of the decision-maker towards a set of possible alternatives X is:

- For any $x, y \in X$, How do we compare x and y ?

1. We prefer x to y ($x \succ y$)

2. We prefer y to x ($y \succ x$)

or, 3. We are indifferent ($x \sim y$)

Basically, Theory of consumer preference is based on ~~three~~ ^{four} assumptions. These are:

1. Completeness

2. Transitivity ~~and~~

3. Non-satiation . and

4. Reflexivity.

Completeness:

For any pair of alternatives $x, y \in X$, the individual decision maker:

$\rightarrow x \succ y$, or

$\rightarrow y \succ x$, or

$\rightarrow x \sim y$.

The decision maker is allowed to choose one and only one of the above options.

Which can more precisely be written as:

either, $\sum_{i=1}^N x_i \geq \sum_{i=1}^N y_i$ (which implies $x \succeq y$) or,

$\rightarrow \sum_{i=1}^N y_i \geq \sum_{i=1}^N x_i$ (which implies $y \succeq x$) or,

both, $\sum_{i=1}^N x_i = \sum_{i=1}^N y_i$ (which implies $x \sim y$)

Reflexivity: Every alternative x is weakly preferred to, at least, one alternative: itself.
A preference relation satisfies reflexivity if, for ~~every~~ any alternative $x \in X$, we have that:

1) $x \sim x$: any bundle is indifferent to it

2) $x \geq x$: any bundle is preferred or indifferent to itself

3) $x \not\sim x$: any bundle belongs to at least one indifference set, namely, the set containing itself if nothing else.

Which can more precisely be written as:

$$\text{If } x \geq y, \quad \sum_{i=1}^N x_i \geq \sum_{i=1}^N y_i \quad \text{and}$$

$$\text{if } y \geq z, \quad \sum_{i=1}^N y_i \geq \sum_{i=1}^N z_i$$

Then it must be that,

$$\sum_{i=1}^N x_i \geq \sum_{i=1}^N z_i \quad (\text{which implies } x \geq z, \text{ as required})$$

For example, $\text{apple} \geq \text{banana}$ and $\text{banana} \geq \text{orange}$

So, $\text{apple} \geq \text{orange}$ (by transitivity)

Rationality: The preference relation $\succeq$ is rational if it possesses the following two properties:

① completeness: for all $x, y \in X$,
either, $x \succeq y$, or, $y \succeq x$, or both.

② Transitivity: for all $x, y, z \in X$,
if $x \succeq y$ and $y \succeq z$, then it must be that
 $x \succeq z$.

[Since 12 total assumption is 2
explanation and interpretation
(6+6)]

Question - 2: Define utility function. Explain the desirability of consumer's preference by the axiom of monotonicity and strong monotonicity.

Utility function: A function $u: X \rightarrow \mathbb{R}$ is a utility function representing preference relations $\succeq$, if, for every pair of alternatives $x, y \in X$.

That is,

$$x \succeq y \Leftrightarrow u(x) \geq u(y)$$

There are two points of the utility function.

These are —

① Only the ranking of alternatives matters

That is, it does not matter if —

$$u(x) = 14 \text{ or if, } u(x) = 2000$$

$$\text{and } u(y) = 10 \text{ or if, } u(y) = 3$$

We do not care about cardinality (the number that the utility function associates with each alternative) but instead care about ordinality (ranking of utility values among alternatives)

② If we apply any strictly increasing function $f(\cdot)$ on $u(x)$, i.e.,

$$f: \mathbb{R} \rightarrow \mathbb{R} \text{ such that } v(x) = f(u(x)).$$

The new function keeps the ranking of alternatives intact and therefore, the new function still

represents the same preference relation.

Example - $v(x) = 3u(x)$
 $v(x) = 5u(x) + 8$

Desirability:

We can express desirability in different ways:

- Monotonicity
- Strong monotonicity
- Non-satiation
- Local non-satiation

Now,

Consider that, x is an n -dimensional bundle

where,

$$x \in \mathbb{R}^n, \text{ i.e., } x = (x_1, x_2, \dots, x_n)$$

where, its k th component represents the amount of good (or, service) k , $x_k \in \mathbb{R}_+$.

Monotonicity: A preference relation satisfies

monotonicity if, for all $x, y \in X$,

where $x \neq y$;

Then the circumstances would be:

- (a) $x_k \geq y_k$ for every good k implies $x \succeq y$
- (b) $x_k \geq y_k$ for every good k implies $x > y$
- $x_k > y_k$ ←

That is,

→ Increasing the amounts of some commodities (without reducing the amount of any other commodity) cannot hurt, $x \succeq y$; and

→ Increasing the amounts of all commodities is strictly preferred, $x > y$.

Strong Monotonicity:

A preference relation satisfies strong monotonicity if, for all $x, y \in B$ where $x \neq y$; then the situation would be for strong monotonicity:

$x_k \geq y_k$ for every good k implies $x > y$
and $x_l > y_l$ for at least one good l .

That is; even if we increase the amounts of only one of the commodities, we make the consumer strictly better off.

[N.B.] Relationship between monotonicity and utility function:

Monotonicity in preferences implies that the utility function is weakly monotonic (weakly increasing) in its arguments.

That is, 'increasing some of its arguments weakly increases the value of the utility function, and increasing all its arguments strictly increases its value.'

For any scalar $\alpha > 1$,

$$U(\alpha x_1, x_2) \geq U(x_1, x_2)$$

$$U(\alpha x_1, \alpha x_2) > U(x_1, x_2)$$

Relationship between strong monotonicity and utility function:

Strong monotonicity in preferences implies that the utility function is strictly monotonic (strictly

increasing) in all its arguments.

That is, increasing some of its arguments strictly increases the value of the utility function.

For any scalar, $\alpha > 1$,

$$U(\alpha x_1, x_2) > U(x_1, x_2)$$

Question - 3:

Show that,

Strong Monotonicity $\Rightarrow$ Monotonicity

We can express desirability in two different ways. These are -

1. Monotonicity
2. Strong monotonicity.

Consider that, x is an n -dimensional bundle.

$$x \in \mathbb{R}^n, \text{ i.e., } x = (x_1, x_2, \dots, x_n)$$

Where, its ~~both~~ k th component represents

the amount of good (or service) k , $x_k \in \mathbb{R}_+$.

Monotonicity: A preference relation satisfies monotonicity if, for all $x, y \in X$, where $x \neq y$,

- a) $x_k \geq y_k$ for every good k implies $x \succeq y$.
- b) $x_k > y_k$ for every good k implies $x \succ y$.

That is,

→ Increasing the amounts of some commodities (without reducing the amount of any other commodity) cannot hurt, $x \succeq y$; and

→ Increasing the amounts of all commodities is strictly preferred, $x \succ y$.

Relationship between monotonicity and utility function:

Monotonicity in preferences implies that the utility function is weakly monotonic (weakly increasing) in its arguments.

That is, increasing some of its arguments weakly increases the value of the utility function,

and increasing all its arguments strictly increases its value.

Like that, For any scalar $\alpha > 1$,

$$U(\alpha x_1, x_2) \geq U(x_1, x_2)$$

$$U(\alpha x_1, \alpha x_2) > U(x_1, x_2)$$

For example:

$$U(x_1, x_2) = \min \{x_1, x_2\}$$

$\Rightarrow$ Monotone, since

$$\min \{x_1 + \delta, x_2 + \delta\} > \min \{x_1, x_2\}$$

For all $\delta > 0$

But not strongly monotone, since

$$\min \{x_1 + \delta, x_2\} \not> \min \{x_1, x_2\}$$

$$\text{if } \min \{x_1, x_2\} = x_2$$

Strong Monotonicity:

A preference relation satisfies strong monotonicity if, for all

$x, y \in X$, where $x \neq y$,

$x_k \geq y_k$ for every good k implies $x \succ y$.
and $x_l \geq y_l$ for at least one good l .

That is, even if we increase the amounts of only one of the commodities, we make the consumer strictly better off.

Relationship between strong monotonicity and utility function:

Strong monotonicity in preferences implies that the utility function is strictly monotonic (strictly increasing) in all its arguments.

That is, increasing some of its arguments strictly increases the value of the utility function.

Like that, for any scalar, $\alpha > 1$,

$$U(\alpha x_1, x_2) > U(x_1, x_2)$$

For example, $U(x_1, x_2) = x_1 + x_2$

This is monotonic, since

$$(x_1 + \delta) + (x_2 + \delta) > x_1 + x_2 \text{ for all } \delta$$

This is also strongly monotonic, since

$$(x_1 + \delta) + x_2 > x_1 + x_2$$

Hence, from the above analysis we can say that, strong monotonicity implies monotonicity but the converse is not necessarily true.

That is, Strong Monotonicity $\not\Rightarrow$ Monotonicity [shown]

Question-4: Show that,

Strong Monotonicity $\not\Rightarrow$ Monotonicity $\not\Rightarrow$ LNS

Answer-3 +

Non-satiation (NS): A preference relation

satisfies NS if, for every $x \in X$, there is another bundle in set X , $y \in X$, which is strictly preferred to x , i.e., $y \succ x$.

Non-satiation is too general, since we could think about a bundle, y , containing extremely large amounts of some goods than x .

then, how far away are y and x ?

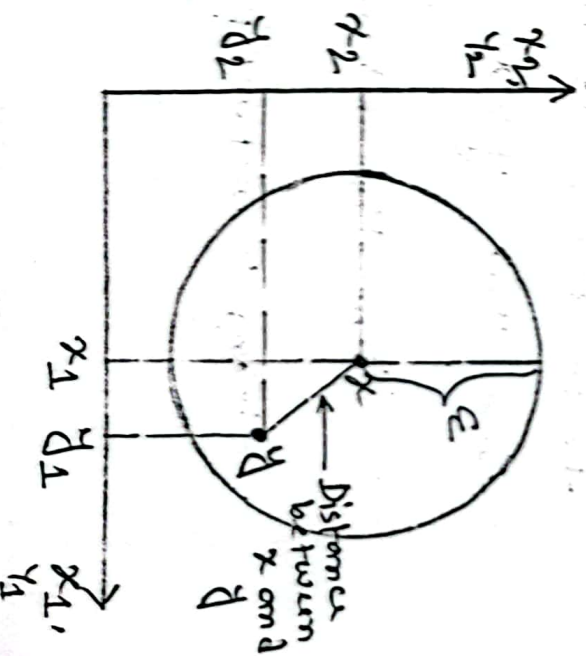
Local non-satiation (LNS):

A preference relation satisfies LNS if, for every bundle $x \in X$ and every $\varepsilon > 0$, there is another bundle $y \in X$ which is less than ε -away from x , $\|y - x\| < \varepsilon$, and for which $y \succ x$.

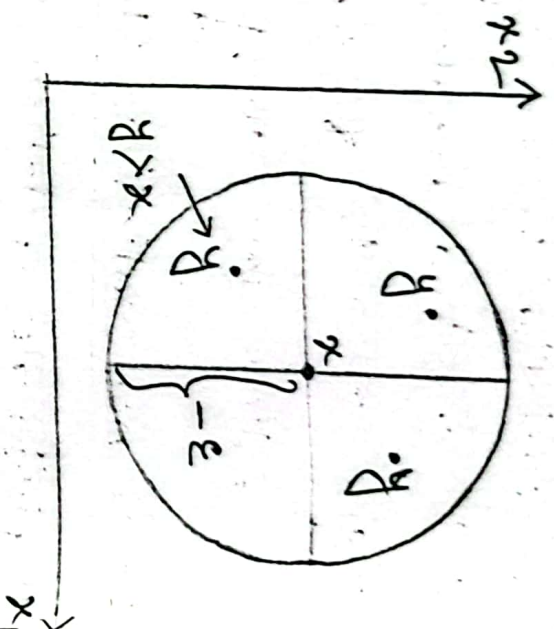
Here,
 $\rightarrow \|y - x\| = \sqrt{(y_1 - x_1)^2 + (y_2 - x_2)^2}$ is the Euclidean distance between x and y , where $x, y \in \mathbb{R}_+^2$.

In words, for every bundle x , and for every distance, ε from x , we can find a more preferred bundle, y .

A preference relation satisfies $y \succ x$ even if bundle y contains less of good-2 (but more of good-1) than bundle, x .



A preference relation satisfies $y \succ x$ even if bundle, y contains less of both goods than bundle, x .



Thus, if a preference relation satisfies monotonicity it must also satisfy Local-non-satiation (LNS).

Given a bundle,

$x = (x_1, x_2)$, increasing all of its component yields a bundle $(x_1 + \delta, x_2 + \delta)$, which is strictly preferred to bundle, (x_1, x_2) by monotonicity.

Hence, there is a bundle $y = (x_1 + \delta, x_2 + \delta)$ such that $y \succ x$ and $\|y - x\| < \epsilon$.

So, from the above analysis we can say that

Strong Monotonicity $\Rightarrow$ Monotonicity $\Rightarrow$ LNS

[shown]

Question-5: Define indifference set. Explain how does strong monotonicity and indifference curve are related?

The indifference set of a bundle $x \in X$ is the set of all bundles $y \in X$, such that $y \sim x$.

$$IND(x) = \{y \in X : y \sim x\}$$

The upper-contour set of bundle x is the set of all bundles $y \in X$, such that $y \succeq x$.

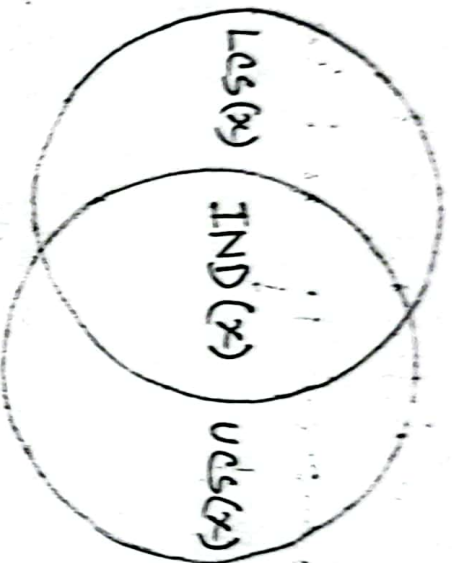
$$UCS(x) = \{y \in X : y \succeq x\}$$

The lower-contour set of bundle x is the set of all bundles $y \in X$, such that $x \succeq y$.

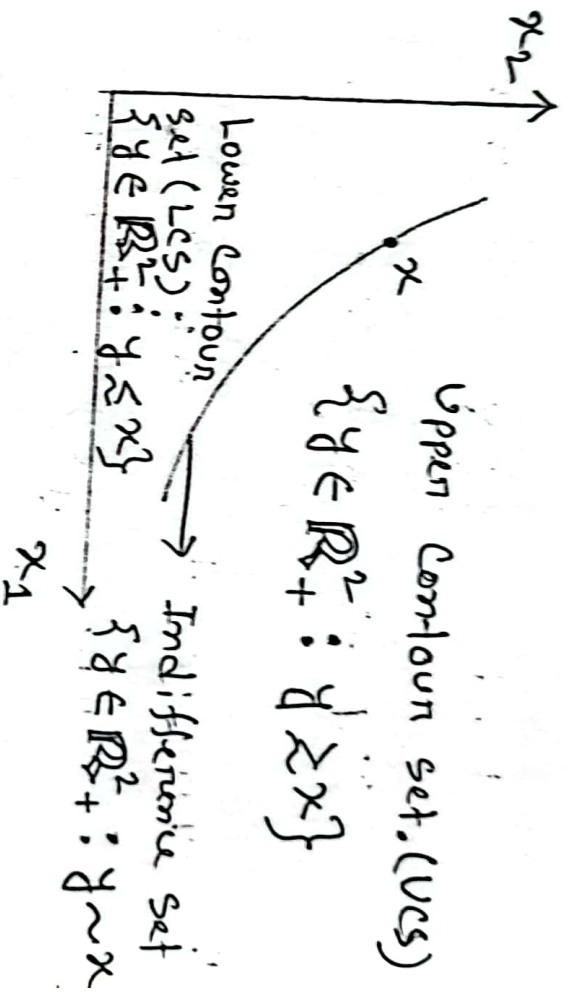
$$LCS(x) = \{y \in X : x \succeq y\}$$

Therefore, $IND(x)$ is the intersection of $UCS(x)$ and $LCS(x)$, that is,

$$IND(x) = UCS(x) \cap LCS(x)$$



That is,

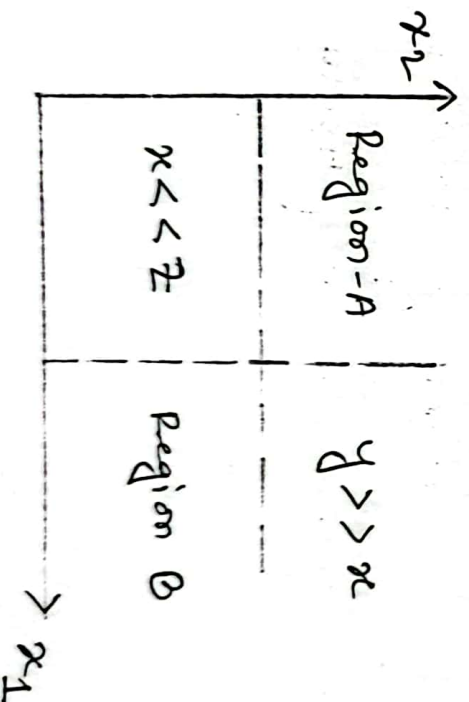


Strong monotonicity implies that indifference curves must be negatively sloped.

To see this, consider a bundle $x \in X$. By strong monotonicity, other bundles $y \in X$, where $y \neq x$, that contain strictly more of all goods than bundle x does, that is, $y \gg x$, must be strictly preferred.

to x , $y \succ x$, thus implying that $y \notin \text{IND}(x)$. Similarly bundle $z \in X$, where $z \neq x$, containing strictly less of every good than bundle x does, that is, $z \ll x$, must satisfy $z \prec x$, thus entailing that $z \notin \text{IND}(x)$.

So for a given bundle $x \in \mathbb{R}_+^2$, bundle satisfying $y \succ x$ lie to the northeast of x and bundle satisfying $z \ll x$ lie to the southwest of x . Since the indifference set, $\text{IND}(x)$ cannot lie on either of these regions by strong monotonicity.



Thus, from the above figure, we can see that the only regions where $\text{IND}(x)$ can lie are regions A and B, which yield a downward

sloping indifference curve passing through x .

Alternatively, for indifference curves to be upward sloping, the preferences cannot satisfy strong monotonicity, but they can satisfy a milder form of desirability such as local non-satiation (LNS).

Thus, from the above analysis, we can say that -

Strong monotonicity implies that indifference curves must be negatively sloped.

Question-6: What are the two definitions of convexity? In what respect they are different from strong convexity? Examine graphically.

Generally, we use two equivalent definitions of for convexity of preferences.

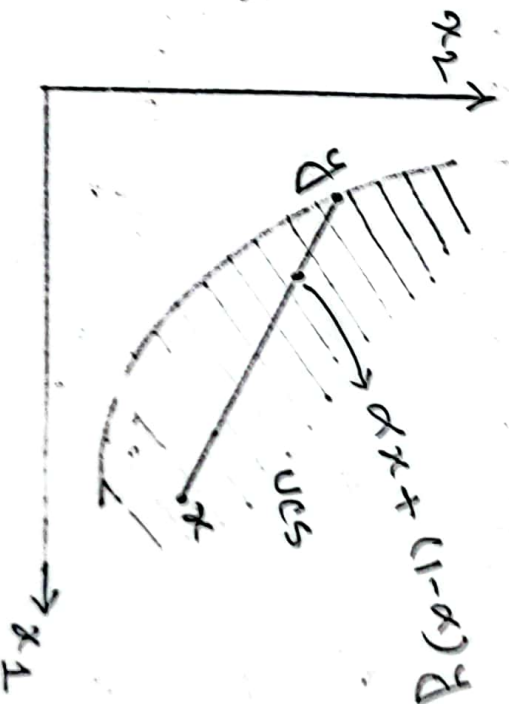
Convexity-1: A preference relation satisfies convexity.

if, for all $x, y \in X$,

$$x \succeq y \implies \alpha x + (1-\alpha)y \succeq y$$

for all $\alpha \in (0, 1)$.

Convexity-1:



Intuitively, the convex combination of bundles x and y is weakly preferred to bundle y alone. The above figure depicts a convex preference

relation sphere, for simplicity, bundle x lies in the upper - contour set of bundle $y, x \in ucs(y)$.

Convexity-2:

A preference relation satisfies convexity if, for every bundle x , its upper contour set is convex.

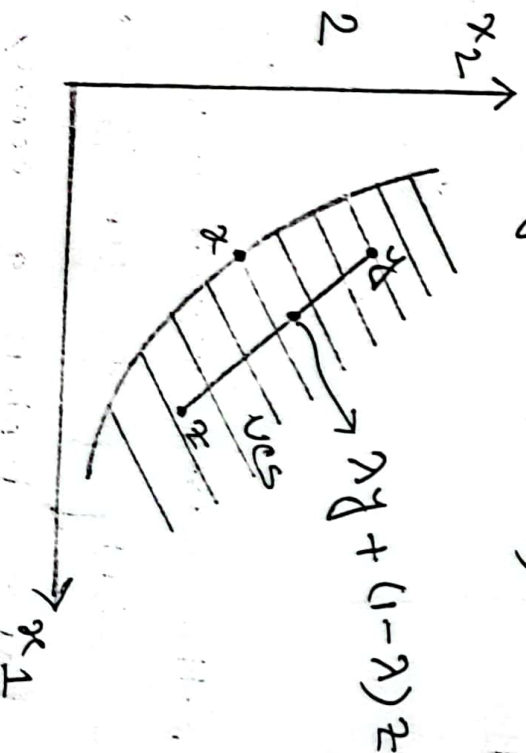
$$ucs(x) = \{y \in X : y \succeq x\} \text{ is convex}$$

That is, for every two bundles y and z ,

$$\begin{cases} y \succeq x \\ z \succeq x \end{cases} \Rightarrow \lambda y + (1-\lambda)z \succeq x$$

for any $\lambda \in (0, 1)$

Convexity-2



This figure shows that, for every two bundles y and z weakly preferred to x , $y \succeq x$ and $z \succeq x$ (which then belong to the UCS of x). Their convex combination:

$\alpha y + (1-\alpha)z$ is also weakly preferred to x :
 $\alpha y + (1-\alpha)z \succeq x$ for any $\alpha \in (0,1)$

Hence, points y, z , and their convex combination belong to the UCS of x .

Strict convexity: A preference relation satisfies strict convexity if, for every $x, y \in X$ where $x \neq y$,

$$\begin{cases} x \succeq z \\ y \succeq z \end{cases} \implies \lambda x + (1-\lambda)y \succ z$$

for all $\lambda \in [0,1]$

We can explain the difference between convexity and strict convexity through graphical representation:

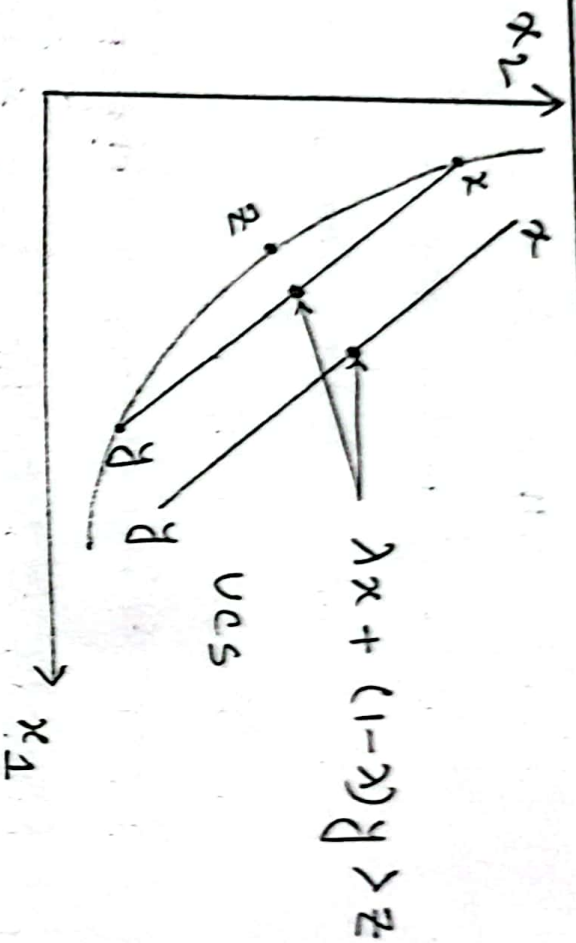


Fig: Strictly Convex Preference

This figure illustrates a preference relation that satisfies strict convexity. When bundle x and y weakly preferred to z , that is, for all $x, y \in X$, where $x \neq y$ such that x and $y \succeq z$, their convex combination $\alpha x + (1-\alpha)y$ where $\alpha \in (0, 1)$, is strictly preferred to bundle z .

↳ Convex but not strict convex preferences:

$$\lambda x + (1-\lambda)y \sim z$$

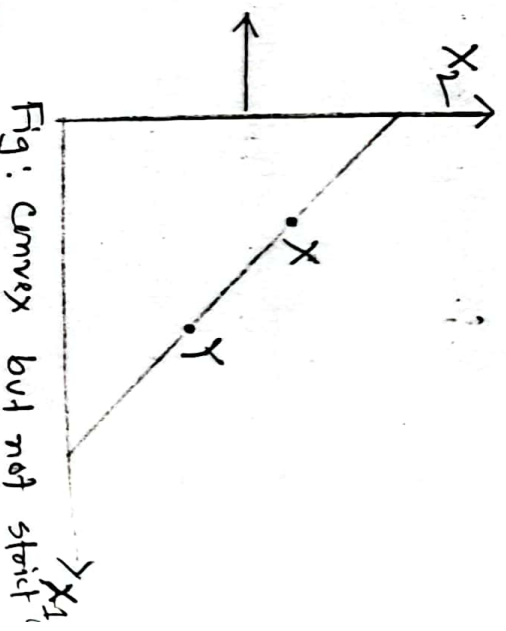


Fig: Convex but not strict

This type of preference relation is represented by linear utility function such as:

$$U(x_1, x_2) = ax_1 + bx_2$$

where x_1 and x_2 are regarded as substitutes.

Another example:

If a preference relation is represented by utility function

such as:

$$U(x_1, x_2) = \min \{ax_1, bx_2\}$$

where, $a, b > 0$, then

the preference relation

satisfies convexity, but

not strict convexity.

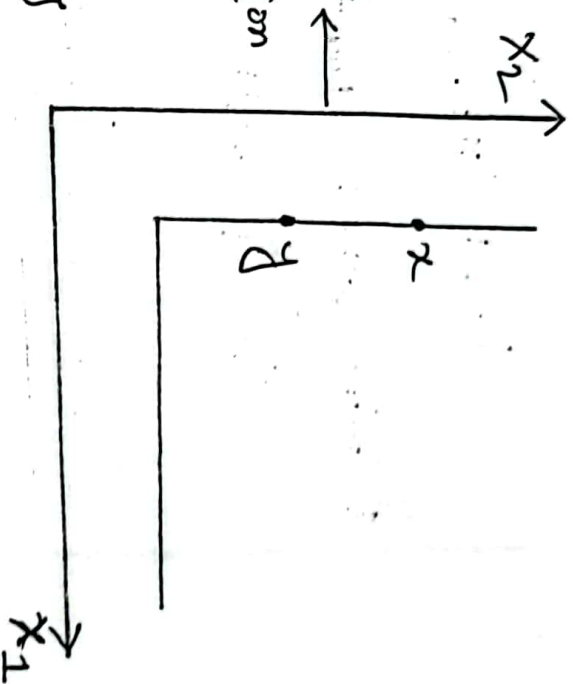


Fig: convex but not strict convex preferences

preference

So, we can say that strict convexity is different from convexity preference.

Question-7: Consider the following utility for

a. $U(x_1, x_2) = ax_1 + bx_2$

b. $U(x_1, x_2) = \min(ax_1, bx_2)$

c. $U(x_1, x_2) = ax_1^{1/2} + bx_2^{1/2}$

d. $U(x_1, x_2) = ax_1^2 + bx_2^2$

Examine which function satisfies convexity, strict convexity or both.

$U(x_1, x_2)$	Satisfies Convexity	Satisfy Strict Convexity
$ax_1 + bx_2$	✓	✗
$\min(ax_1, bx_2)$	✓	✗
$ax_1^{1/2} + bx_2^{1/2}$	✓	✓
$ax_1^2 + bx_2^2$	✗	✗
$ax_1^{1/2} \times bx_2^{1/2}$	✓	✓
$ax_1^2 \times bx_2^2$	✗	✗

We know that, in two dimensions, if indifference curves are straight lines, then preferences are convex, but not strictly convex.

Geometrically, convexity means that the line segment between two points on the graph of f lies on or above the graph itself.

Strict convexity means that the line segment lies strictly above the graph of f , except at the segment's endpoints.

A preference relation that satisfies strict ~~convexity~~ convexity,

when bundles x and y lie on both the upper contour set (UCS) and the indifference curve, IND, the definition of strict convexity is satisfied because the convex combination of these two bundles lies strictly 'inside' the UCS, thus implying $\alpha x + (1-\alpha)y \succ z$.

A preference relation that satisfies convex but not strict convexity:

The convex combination of bundles x and y lies strictly on the indifference curve, $\therefore$ their convex combination therefore produces a bundle $\alpha x + (1-\alpha)y$ that the consumer regards as indifferent to bundle z .

This preference relation is represented by a utility function such as -

$$U(x_1, x_2) = ax_1 + bx_2, \text{ where } a, b > 0$$

which indicates that good 1 and 2 are regarded as substitutes.

For another example,

a preference relation that satisfies convex but not strict convexity, consider preferences over two goods that are regarded as perfect complement, represented by the utility f

$$U(x_1, x_2) = \min \{ax_1, bx_2\}$$

where, $a, b > 0$

Note that, the convex combination of bundles x and y produces a bundle that does not necessarily lie strictly within the VCS.

The above table shows some types of utility functions representing preference relations that may or may not satisfy convexity and strict convexity.

If a preference relation satisfies strict convexity, it must also satisfy convexity, whereas the converse is not necessarily true.

For the utility function:

$$U(x_1, x_2) = ax_1 + bx_2$$

This utility function satisfies convexity but does not satisfy strict convexity.

Similarly, $U(x_1, x_2) = \min(ax_1 + bx_2)$

This also satisfies convexity but does not satisfy strict convexity.

For $U(x_1, x_2) = ax_1^{\frac{1}{2}} + bx_2^{\frac{1}{2}}$

This preference satisfies both convexity or strict convexity.

$$\text{Ex } U(x_1, x_2) = ax_1^2 + bx_2^2$$

This utility function does not satisfy or strict convexity.

$$\text{Ex } U(x_1, x_2) = ax_1^{1/2} \times bx_2^{1/2}$$

This utility function satisfies both convex and strict convexity.

$$\text{Ex } U(x_1, x_2) = ~~ax_1^2~~ \times bx_2^2$$

This utility function also does not satisfy convexity or strict convexity.

Question - 8:

Show that

Convexity of preferences

$\Leftrightarrow$

$U^C(x)$ is convex

$\Leftrightarrow$

$U(\cdot)$ is quasi-concave

A utility function $U(\cdot)$ is quasi-concave if, for any bundle x , the set of all bundles for which the consumer experiences a higher utility, that is,

$$U^C(x) = \{y \in X \mid U(y) \geq U(x)\}, \text{ is convex.}$$

Hence the following three properties are equivalent:

$$\text{Convexity of preferences} \Leftrightarrow U^C(x) \text{ is convex} \Leftrightarrow U(\cdot) \text{ is quasi-concave.}$$

Alternatively, quasi-concavity can be understood as follows:

For every two bundles $x, y \in X$, the utility of consuming the convex combination of these two bundles, $U(\alpha x + (1-\alpha)y)$, is weakly higher than the minimal utility from consuming each bundle separately,

Thus; convexity of preferences $\Leftrightarrow USC(x)$ is $\Leftrightarrow U(\cdot)$ is quasi-concave.
 [showed]

$$\min \{U(x), U(y)\}, \text{ or more compactly, } U(\alpha x + (1-\alpha)y) \geq \min \{U(x), U(y)\}$$

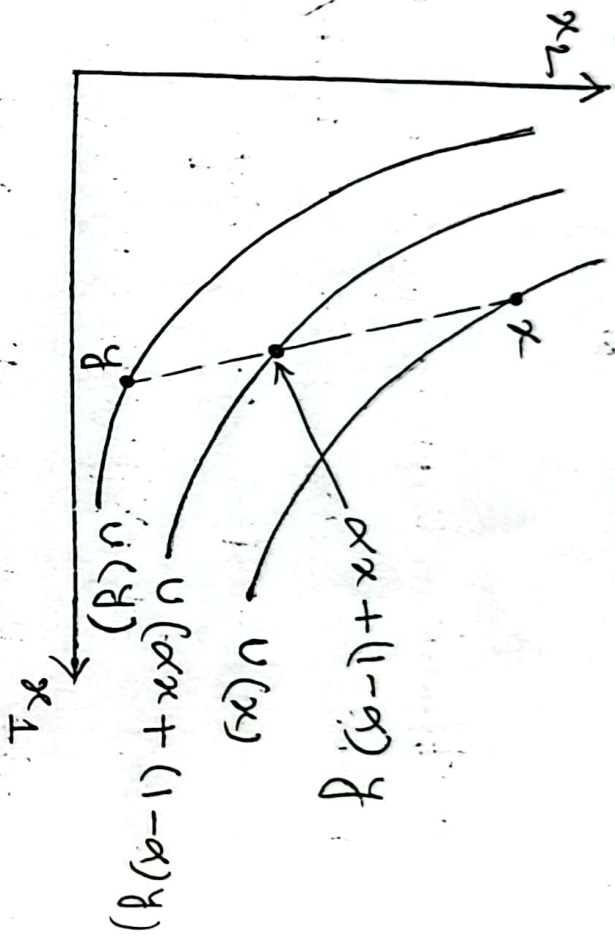


Fig: Quasi-Concavity

This figure illustrates the definition above.

In particular, starting from bundles x and y with associated utility levels $U(x)$ and $U(y)$ respectively, we can construct a convex combination of the two bundles. If the utility that the consumer derives from consuming this convex combination is larger than from consuming the bundle associated to the lowest utility level (which in this case is bundle y we say that this individual's utility function satisfies quasi-concavity.